

Apriori Algorithm

Definition: Apriori Algorithm is an unsupervised machine learning algorithm which is mainly used for association rule learning.

- It uses frequent itemset to generate combinations association rules ~~and it~~.
- It is designed to work on databases that contain transaction.
- * This algorithm uses a BFS (Breadth First Search) and Hash Tree to calculate the itemset associations efficiently.

Frequent Itemset (FI) = FI are those items whose support is greater than threshold value.

eg: $A = \{1, 2, 3, 4, 5, 6\}$ $R = \{1, 2, 3, 7\}$

∴ 2, 3 are FI.

Algorithm Steps

- Step 1: Determine the support of itemset and select min support and confidence.
- Step 2: Take all support in transaction with higher support value than min or selected support value.
- Step 3: Find all rules of these sub itemset that have higher confidence value than threshold or min confidence.

Step ④ Sort the rules in ascending decreasing orders of lift.

Nonnumerical Example

Tid Itemset

T₁ A, B

T₂ B, D

T₃ B, C

T₄ A, B, D

T₅ A, C

T₆ B, C

T₇ A, C

T₈ A, B, C, E

T₉ A, B, C

Assume %

min-support = 2

min-confidence = 50%

Step ① calculating C_1 and L_1

where, L_1 is Frequent itemset

C_1 is candidate set

C_1 :

Itemset	Support count
A	6
B	7
C	5
D	2
E	1

candidate set or C_1

L_1 :

Itemset	Support count
A	6
B	7
C	5
D	2

Here in L_1 , B is excluded and rest A, C and D are kept because min support is 2 and only A, C , and D has support count greater than 2, so, B is excluded.

Step 2: Candidate Generation C_2 and L_2 :

	Itemset	Support-count
C_2	$\{A, B\}$	4
	$\{A, C\}$	4
	$\{A, D\}$	1
	$\{B, C\}$	4
	$\{B, D\}$	2
	$\{C, D\}$	0

	Itemset	Support-count
L_2	$\{A, B\}$	4
	$\{A, C\}$	4
	$\{A, D\}$	1
	$\{B, C\}$	4
	$\{B, D\}$	2

Here, $\{A, B\}$ and $\{C, D\}$ has ≥ 2 support count so excluded.

Step 3: Candidate generation C_3 and L_3

	Itemset	Support count
C_3	$\{A, B, C\}$	2
	$\{A, B, D\}$	0
	$\{B, C, D\}$	0
	$\{A, C, D\}$	0

	Itemset	Support count
L_3	$\{A, B, C\}$	2
	$\{A, B, D\}$	

Here, $\{A, B, D\}$, $\{B, C, D\}$ and $\{A, C, D\}$ are excluded since its support count is < 2 .

Step 4: Finding association rule for sunset.
1st calculate the confidence using
$$\frac{\text{sup}(A \wedge B)}{\text{sup}(A)}$$

∴ sup → support

→ After calculating we will exclude rule that have less confidence than min. threshold (50%).

rule	support	confidence
$A \wedge B \rightarrow C$	2	$\frac{\text{sup}((A \wedge B) \wedge C)}{\text{sup}(A \wedge B)} = \frac{2}{4} = 50\%$
$B \wedge C \rightarrow A$	2	$\frac{\text{sup}((B \wedge C) \wedge A)}{\text{sup}(B \wedge C)} = \frac{2}{4} = 50\%$
$A \wedge C \rightarrow B$	2	$\frac{\text{sup}((A \wedge C) \wedge B)}{\text{sup}(A \wedge C)} = \frac{2}{4} = 50\%$
$C \rightarrow A \wedge B$	2	$\frac{\text{sup}(C \wedge (A \wedge B))}{\text{sup}(C)} = \frac{2}{5} = 40\%$
$A \rightarrow B \wedge C$	2	$\frac{\text{sup}(A \wedge (B \wedge C))}{\text{sup}(A)} = \frac{2}{6} = 33.33\%$
$B \rightarrow A \wedge C$	2	$\frac{\text{sup}(B \wedge (A \wedge C))}{\text{sup}(B)} = \frac{2}{7} = 28\%$

So, the first 3 rules $A \wedge B \rightarrow C$, $B \wedge C \rightarrow A$, and $A \wedge C \rightarrow B$ can be considered as the strong association rule for the given problem since its confidence % is greater than 50%.